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ITAI-2373-Natural Language Processing

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NewsBot Intelligence System – Reflective Journal

Collaboration and Project Management

I completed this project by myself because no groups were assigned. At first, the project looked overwhelming because it included many NLP techniques, but I broke it into smaller tasks and completed one section at a time. I started loading the dataset and preprocessing the text before moving to TF-IDF, sentiment analysis, named entity recognition, and machine learning. I tested each section before connecting everything together. Whenever I found an error, I reviewed my code, fixed the problem, and tested it again. Working through these challenges helped me understand how each part of the system worked.

Technical Integration Challenges

The hardest part was getting every NLP component to work together. Sometimes one change causes another section to stop working because of missing variables or feature size problems. I learned that careful testing is just as important as writing the code. After fixing these issues, I understood much better how an end-to-end NLP pipeline works.

Business Value Assessment

This project showed me how NLP can help businesses organize and analyze large amounts of news quickly. The system can classify articles, analyze sentiments, and identify important

people, organizations, and places. These features could help media companies, businesses, and researchers save time and find useful information faster.

Individual Contributions

I completed every part of this project independently. I prepared the dataset, cleaned the text, created TF-IDF features, performed sentiment analysis, extracted named entities, trained multiple machine learning models, evaluated their performance, and integrated everything into one NewsBot system. I also tested the completed system using new articles.

Future Enhancements

In the future I would like to improve this project by using newer language models such as BERT, adding multilingual support, and creating automatic summaries of articles. Using more training data and adding real-time news analysis could also improve the system.

Professional Development Impact

This project improved my understanding of NLP and machine learning. It also made me more confident using Python, pandas, spaCy, NLTK, and scikit-learn. I learned a lot from debugging my own code and solving problems throughout the project. I plan to include this project in my portfolio because it demonstrates the practical skills I developed during this course.